Can Rapid Transit Improve Democratic Participation? Evidence from Philadelphia Elections (2021-2025)

1 Abstract

Background: Disparities in access to public transportation have perpetuated long-standing inequities in education, social mobility, and public health across urban areas in the United States. However, the relationship between public transportation accessibility and political engagement remains less clearly defined.

Objective: This study investigates the relationship between rapid public transit availability and in-person election day voter turnout across Philadelphia divisions. It further examines how socioeconomic disadvantage, as operationalized by poverty, moderates this relationship.

Methods: The City of Philadelphia's elections data provided the number of in-person votes for each division (N=1703) in each general and primary election (N=9) between May 2021 and May 2025. Rapid transit station locations were spatially linked to each division, the geographic unit in which residents vote. The approximate poverty rate of each division was calculated based on census tract level American Community Survey 2023 5-Year Estimates; in turn, a standardized poverty index was constructed. Poisson regression with clustered standard errors assessed the association between rapid public transit availability and in-person election day voter turnout as well as how the standardized poverty level in each division affected this relationship.

Results: The interaction terms indicate that the negative relationship between poverty and turnout is stronger in divisions with limited or no rapid transit access. In low-moderate transit divisions, each one-unit increase in the poverty score is associated with an additional 8% decrease in expected turnout relative to what would be expected in a high-transit division at the same poverty level. In divisions with no rapid transit, this effect is 13.6%.

Conclusion: Rapid public transit availability may mitigate the negative effects of poverty on in-person voter turnout. These results underscore how accessible public transit can promote political representation of vulnerable groups, mobilize urban voters in swing states, and act as a tool for social mobility.

2 Introduction

Public transportation facilitates the movement of people and goods, enabling efficient urban systems. Transportation research has consistently demonstrated that strong public transportation networks have contributed to improved environmental and health outcomes by reducing emissions associated with personal automobile use. Additionally, strong public transportation networks promote public health through increased walkability and higher air quality. However, the benefits associated with public transportation extend beyond environmental and health outcomes, connecting residents clustered by deeply ingrained social and economic disparities.

While the social, economic, and environmental consequences of insufficient public transportation access are well documented, there is less consensus surrounding its implications for democratic participation. Urban environments in the United States are characterized by high levels of demographic diversity and pronounced social inequalities that necessitate political representation. Thus, declining levels of democratic participation in urban areas pose challenges to the effective functioning of American democracy. A democratic system depends on civic engagement across diverse populations, a requirement that is particularly salient in urban contexts that are home to disproportionately vulnerable groups, including lower-income households.

In this project, I explored the relationship between public transportation access and political participation, as well as how economic disparities further influence this relationship. Specifically, this project quantitatively assessed the relationship between rapid transit availability and in-person voter turnout in Philadelphia from 2021-2025 at the division level. Furthermore, it assessed how socioeconomic disadvantage, as operationalized by poverty, moderates this relationship.

3 Literature Review

3.1 Public Transit and Inequality

Across urban environments in the U.S., disparities in access to public transportation have led to spatial inequalities within the built environment, negatively impacting social equity, including access to educational and employment opportunities (Grengs, 2010). Although easy access to public transportation has become a key trait of urban areas, the availability of public transportation varies throughout these environments. Lower-income and Black and Hispanic communities tend to rely more upon public transportation, nonetheless those communities often reside in urban transit deserts that disconnect them from economic and social opportunities (Anderson, 2016).

These dynamics have triggered long-term imbalances in access to opportunities across urban landscapes. During the postwar period, patterns of white flight from the inner cities into the suburbs shifted the racial and ethnic composition of American cities, with primarily lower-class and racial minority groups remaining. These patterns of white flight have played a pivotal role in shifting the location of employment opportunities. The "spatial mismatch hypothesis" (SMH) argues that the spatial distribution of high-skill employment opportunities shifted out of the inner cities and into the suburbs during the postwar period. Lower-skilled racial and ethnic minorities residing in the inner cities were left unable to access better employment opportunities due to the spatial disconnect and barriers to access via public transportation availability (Kain, 1969). SMH accounts for how previously existing racial inequalities between the inner cities and suburbs are reinforced by unequal transportation access, with Black and Hispanic

communities bearing the brunt of this inaccess. Such urban planning decisions not only generate economic exclusion, but also political marginalization by shaping who can readily access civic and political institutions (Trounstine, 2018).

In the educational sector, the availability of public transportation has also played a pivotal role in facilitating better access to quality public education. In inner city areas that implement school choice programs, allowing students to attend magnet schools, students often commute via public transportation, walking, and biking (McDonald, 2008). This and other studies show that inner city students rely upon public transportation to attend magnet schools (Scott & Marshall, 2019). Additionally, studies have found that transportation access increases the attendance and decreases the rate of chronic absences among economically-disadvantaged students (Edwards, 2023).

Altogether, this literature establishes that public transportation access is crucial for enabling social mobility through access to employment opportunities and education. Insufficient access to public transportation limits access to these social institutions, exacerbating pre-existing socioeconomic inequalities in urban environments.

3.2 Transportation and Cost of Voting

Political participation in urban environments is essential to a functioning democracy in which residents maintain the agency to engage with the political institutions that shape their everyday lives. In comparison to other areas of transportation research, there is less literature coming to a consensus on how public transportation access influences an individual's propensity to turnout on election day.

Political scientists often model voter turnout through a rational choice framework. Downs' (1957) *An Economic Theory of Democracy* proposes that individuals vote when the perceived utility outweighs the costs, as modeled by the equation $R = P(B) - C$. Downs' Rational Voter Theory utilizes economic principles to model how individual voters act as rational actors and different political parties operate as competing actors, similar to competing firms (Downs, 1957). R refers to the expected reward from voting, P refers to the probability that one's vote will change the outcome, B refers to the benefit of one's preferred candidate winning, and C refers to the cost of voting. According to this model, an individual will vote if $P(B) > C$. **Thus, I argue that insufficient public transportation access may have embedded costs (C) such as lost wages, time, and childcare, reducing an individual's willingness to turnout**.

Many empirical studies support Downs' Rational Voter Theory, viewing transportation as an inherent cost to voting. The literature reveals that voter turnout is determined by the logistical costs of voting (Brady & McNulty, 2011). Upon looking at California's 2003 governor's election in which the polling location for many voters changed, one study finds that unexpected polling location changes decrease election day turnout. This decrease was observed among less educated, low income, and less politically engaged voters, highlighting how barriers to access for voting that increased the cost (C) can significantly decrease voter turnout among select groups that are more prone to facing such costs. Similarly, unexpected changes in polling locations in California depressed voter turnout, especially among low-income and less educated voters (Haspel & Knotts, 2005). Much of the literature comes to the consensus that factors like time, mobility, and informational access influence disparities in civic engagement (Verba et al., 1995). However, many of these logistical factors are further amplified by overarching institutional barriers and spatial inequalities (Burden et al., 2017).

3.3 Evidence on Transit and Turnout

Research assessing the impacts of public transportation availability on political participation have yielded mixed results, depending on the operationalization of public transportation availability and the specific geographic context.

There is strong evidence to support the idea that there is a link between car ownership and in-person election day turnout. One study investigating the relationship between car ownership and voter turnout in Michigan determined that lack of car access weakens voter day turnout, in which these effects further weakened turnout the farther away a given voter's home address was from their registered polling location (de Benedictis-Kessner & Palmer, 2023). This study not only suggests that an individual's means of transportation, owning a car, impacts turnout, but distance acts as a factor as well.

Building off of the idea of distance as an additional cost to voting, another study suggests that there is a nonlinear relationship between increased distance and an individual's propensity to vote. In the context of French elections, voters have the option to be registered in a different municipality than they reside in, allowing them to travel greater distances when voting in-person. The study found that at the median distance between voter home addresses and their registered polling locations, an increase in distance was associated with lower turnout rates (Fauvelle-Aymar & François, 2018). However, this study also found that an increase in distance for voters that were within a shorter distance of their registered polling location had more profound effects on turnout, indicating that the effects of distance may be more pronounced in circumstances in which individuals are relatively closer to their polling location. Thus, based on this study, though an individual's home distance from their respective polling location influences their propensity to turnout, there may also be diminishing marginal costs to voting as distance increases past a certain threshold. This provides further evidence for distance as a perceived cost of voting, though the relationship may be nonlinear.

Looking specifically at how public transportation influences turnout, the results become more varied depending on the geographical context. In the context of public transportation in the U.S., one study relied on a random sample of 1,500 Pennsylvania voters to determine how proximity to public transit impacts voter turnout in the 2016 and 2018 elections. The findings revealed that greater proximity to public transit provides a modest, yet reliable and statistically significant impact on voter turnout in low-salience elections (Klemick, 2022). Conversely, greater proximity to public transit provides a smaller but still statistically significant impact on voter turnout for high salience elections, in comparison to low salience elections. However, contrary to previously established beliefs, the study found that voter turnout in zip codes with a higher median income tended to be more affected by closer proximity to public transit. This contrasts with previously established notions that lower-income communities are more vulnerable to the cost of voting caused by insufficient transportation access.

In an international context, some studies investigating the role of public transportation on voter turnout suggest that the monetary cost of public transportation is not a significant factor in determining voter turnout. A study by Pereira et al. found that fare-free transit increased ridership between 7.2% and 17.5% on election days during the 2022 Brazilian national elections, but estimated a precise and robust null effect of the policy on voter turnout (Pereira et al., 2023). Factors such as the benefits citizens receive from voting outweighing the costs of transit, compulsory voting policies, and already being in close proximity to polling locations may have contributed to this effect.

3.4 Gaps in Existing Literature

Overall, the aforementioned literature establishes that transportation is a component for shaping access to political participation in the U.S. and other international contexts. Nonetheless, there are still gaps in the literature surrounding how to measure the relationship between public transportation and political participation. Primarily, studies based in the U.S. focus on polling site location and car ownership. However, there is limited research on how public transportation availability affects turnout as measured by access to rapid transit. This study will select a smaller geographic area, a city, as opposed to the state or national level, allowing for a more granular approach in evaluating transportation access and demographic characteristics.

3.5 Philadelphia Context: Demographics, Politics, and Transit

Here, I argue that Philadelphia is a prime location to study the relationship between transit, turnout, and poverty as: (1) Philadelphia is a racially diverse city in which Black and Hispanic residents experience higher rates of poverty, (2) Philadelphia is a key Democratic anchor in Pennsylvania, a swing state, and (3) Philadelphia's long-established public transportation authority is under threat due to insufficient funds caused by the COVID-19 pandemic.

Home to almost 1.6 million residents, Philadelphia is the sixth largest city in the U.S. as of 2025 and is located in the Mid-Atlantic region (The Pew Charitable Trusts, 2025). Given its large population size, Philadelphia encompasses a variety of landscapes and demographic characteristics. In 2023, Philadelphia's population was composed of approximately 39.5% Black, 34.6% White, 15.8% Hispanic/Latino, 8% Asian, 8.9% Multiracial, and .5% Native American people (U.S. Census Bureau, 2023). Despite its racially diverse population, the city's Black and Hispanic residents experience higher rates of poverty and earn significantly less than the city median income (League, 2025).

Alongside these demographic characteristics, Philadelphia election results have key implications for state and national elections. Located in Pennsylvania, a swing state, Philadelphia had approximately 1.033 million registered voters in 2023 (Philadelphia City Commissioners, 2023). Philadelphia makes up around 12% of all eligible voters in the state. Around 3 out of every 4 registered voters in Philadelphia identify as Democrats and only 1 in every 9 identify as Republicans (Philadelphia City Commissioners, 2023). Thus, Philadelphia elections maintain the power to sway state and national election outcomes as it acts as an urban anchor in Pennsylvania for the Democratic party.

Philadelphia has 66 wards, which are further divided into 1,703 divisions. The division an individual resides in determines where they vote as each division contains its own polling location. Following the Pennsylvania election schedule, there are primaries, typically in May, where voters elect their nominees by political party, then followed by general elections in November. Pennsylvania has implemented a closed election system in which registered voters can only vote in the primary of their registered political party (Pennsylvania Department of State, 2023).

Furthermore, Philadelphia's public transportation evolution provides insight on the government's role in facilitating public transportation sustainment and investment. Philadelphia's public transit system has greatly evolved since the nineteenth century. By the end of the nineteenth century, electrified streetcars replaced horse-drawn forms of transportation (Hepp, n.d.), thus began the development of Philadelphia's privately-owned streetcar network. The Philadelphia Rapid Transit Company (PRT) pioneered early subways and elevated lines, eventually evolving into the Philadelphia Transportation Company (PTC). Philadelphia developed the Market Frankford Line (MFL) in 1907, transporting residents from east to west and north to south along Philadelphia's northeastern border (Hutto, 2012). In 1928, the Broad Street Line (BSL) began transporting residents from north to south along Broad Street (Newlin,

2007). By the 1940s, automobile ownership sharply increased, reducing the amount of streetcar ridership (Hepp, n.d.). This contributed to transit companies facing financial difficulties and subsequent public concern in preserving access to public transit. In response, the Pennsylvania General Assembly created the Southeastern Pennsylvania Public Transit Authority (SEPTA) in 1963; its aim was to manage transit operations across the five-county region (Bucks, Chester, Delaware, Montgomery, and Philadelphia). By 1968, SEPTA took over the PTC, managing the operations of the former private transit companies and consolidating streetcar, bus, and subway routes all into one public sector authority. Within Philadelphia, SEPTA currently operates rapid transit, which comprises the Broad Street Line (subway), Market Street Line (subway), and a trolley network. SEPTA also operates high-speed rails, buses, and regional commuter rails.

In recent history, the COVID-19 pandemic, which began in 2020, contributed to decreases in ridership (MacDonald, 2025). While bus and subway ridership quickly rebounded, regional rail ridership remains below its pre-pandemic level. The loss of ridership meant that SEPTA experienced a $240-$320 million annual operating deficit, only kept afloat by emergency federal funds (Council, 2022). Given that the recent SEPTA budget cuts planned to cut 45% of services and increase fares for riders (SEPTA, 2025), it is crucial to investigate how disinvestment in public transit systems may pose consequences for political participation.

Taken together, Philadelphia's demographic diversity, political significance, and public transit history provide a critical context for understanding how patterns of civic engagement are impacted by public transportation availability and ongoing social inequities.

4 Data

This study used data from publicly available voting, public transportation, and demographic data sources. This data was used to investigate the relationship between rapid public transit availability and in-person voter turnout.

To measure in-person voter turnout, this study uses count data at the election-division level from the Philadelphia City Commissioners Office. This data includes five primary elections and four general elections, in which nine total elections occurred between May 2021 and May 2025 (Philadelphia City Commissioners, 2025). The division is the smallest unit used for administering elections in Philadelphia; there are 1,703 divisions in Philadelphia which sit within 66 wards. These data do not reference individual's identities, only the count of registered voters, in-person election day votes, provisional ballots, and mail-in votes per division per election.

To measure rapid public transit availability, this study uses data provided by SEPTA's open data repository (Southeastern Pennsylvania Transportation Authority, 2025), with analyses focusing on the latitude and longitude of physical, stationary rapid transit stations in Philadelphia. There were no new rapid transit developments, in the form of physical stations opening or closing, during the time period of this study (2021-2025) as confirmed by SEPTA.

To measure socioeconomic status, this study uses census-tract level American Community Survey (ACS) 2023 5-Year Estimates from the *tidycensus* package in R to create a standardized poverty score across different divisions in Philadelphia (Walker & Herman, 2025).

Additionally, this analysis uses shapefiles of both census tract (U.S. Census Bureau, 2025) and Philadelphia division boundaries (City of Philadelphia, 2025).

5 Methods

This study assesses the relationship between rapid transit availability and voter turnout moderated by poverty at the division level (N=1703) for a given election (N=9) during the period between 2021 and 2025 in Philadelphia. The unit of observation for this analysis is each division i during election t.

First, to quantify the availability of rapid transit, I generated the count of fixed physical rapid transit stations per division i as a fixed number between 2021-2025. Specifically, I define rapid transit availability as the number of subway and/or trolley stations in a given division. For this analysis, I exclude regional rail stops because this may include trips that involve commutes outside the boundaries of Philadelphia, possibly capturing travel among non-residents. Additionally, this study does not include buses in its count of rapid transit availability, due to the less consistent nature of bus availability and more frequent addition and/or cancellation of routes across years. Thus, in operationalizing rapid transit availability, this study only includes the count of physical subway stations (Market-Frankford line and the Broad Street line) and trolley stations.

I linked every rapid transit station to the division boundaries for Philadelphia. Then, I generated a count of rapid transit stops in each division. For this study, I categorize divisions as having high rapid transit (two or more stops), low-moderate rapid transit (one stop), and no rapid transit (no stops). The map below displays the rapid transit availability by division in Philadelphia, see Figure 1.

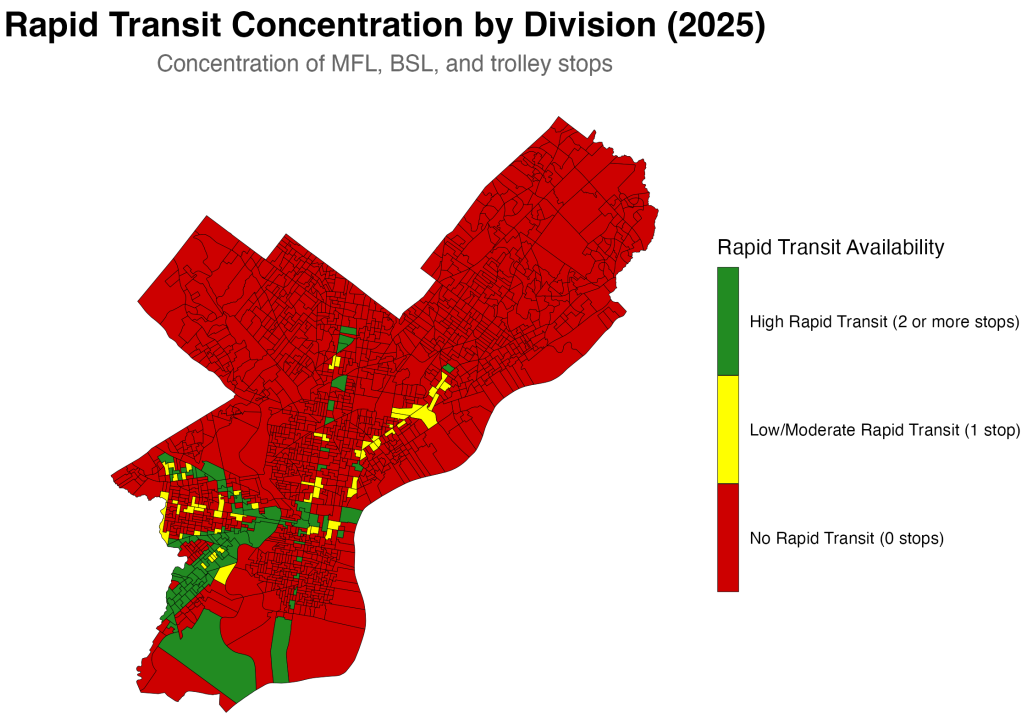


Figure 1: Rapid Transit Availability in Philadelphia by Division, 2025

Next, the in-person election day turnout rate was calculated for each division (N=1703) in each of the nine elections that occurred between May 2021-May 2025. There were five primary elections and four general elections (N=9). For descriptive statistics, I calculated an in-person election day turnout rate per division i in election t, I

divided the total in-person election day votes by registered voters in a given division i for election t.

In addition to gathering in-person election day turnout and rapid public transit data, it was crucial to account for how underlying socioeconomic patterns moderate this relationship between transit and turnout. I used the 2023 ACS 5-Year Estimate of poverty rate for each census tract provided by the *tidycensus* package in R as a measure for the socioeconomic status of a division. A key methodological challenge was that the poverty rate data was only available at the census tract-level but not the division-level. Division boundaries and census tract boundaries are not coterminous. Some divisions had multiple overlapping census tracts. Also, population count data was not available at the division-level.

To calculate the poverty score of each census tract, I determined the average citywide poverty rate. In 2023, the poverty rate of Philadelphia was approximately 21.39%. To capture each census tract's relative poverty rate, I calculated how many standard deviations away each census tract was from the citywide average, creating a standardized z-score for each tract's poverty level. I used weighted means and standard deviations to calculate the z-score; the weights were based on each census tract's share of Philadelphia's total population.

To adjust the geographic unit from the tract-level to the division-level, I implemented a weighting procedure based on land area, rather than population. For example, division 14-01 in Philadelphia is located between three different census tracts–specifically, census tracts 132, 133, and 376. Census tract 132 makes up approximately 60.9% of the division area and has a poverty score of 1.2, census tract 133 makes up approximately 39% of the division area and has a poverty score of -.574, and census tract 376 makes up approximately .0007% of the division area and has a poverty score of -.23. Given this information, the standardized poverty score for division 14-01 is calculated through a weighted average of (.609 * 1.2) + (.39 * -.574) + (.000007 * -.23). Using this weighted calculation, division 14-01 has a poverty z-score of .64. This weighting procedure applies to 1,371 divisions in the data that are not fully captured by one census tract, representing 80.5% of all divisions in Philadelphia. Therefore, the weighting procedure is crucial in ensuring spatial consistency in the unit of measurement for the data used for analysis.

To reiterate, the objective of this study is to investigate the relationship between rapid public transit availability and in-person voter turnout across Philadelphia voting divisions from 2021 to 2025, examining how socioeconomic disadvantage, as operationalized by poverty, moderates this relationship. Although descriptive figures were calculated, these calculations do not account for temporal variations.

$$\begin{aligned}\log\left(\frac{E(Election\ Day\ Votes_{it})}{Registered\ Voters_{it}}\right) = \beta_0 &+ \beta_1\, Poverty\ Score_i + \beta_2\, No\ Transit_i \\ &+ \beta_3\, Low\ Transit_i + \beta_4\,(Poverty\ Score_i\ No\ Transit_i) \\ &+ \beta_5\,(Poverty\ Score_i\ Low\ Transit_i) + a_t\end{aligned}$$

A Poisson regression model was used to analyze the association between rapid public transit availability and in-person voter turnout across Philadelphia divisions. The model also tested how varying levels of poverty affected this relationship. Here, $E(Election\ Day\ Votes_{it})$ is the expected number of people who voted in division i in election t. The resulting ratio $\frac{E(Election\ Day\ Votes_{it})}{Registered\ Voters_{it}}$ captures the in-person election day turnout rate — the expected number of votes cast per registered voter in division i during election t. The poverty score coefficient (β_1), when exponentiated, provides the expected multiplicative change in turnout rate for every one unit increase in standardized poverty score in division i. The no transit coefficient (β_2), when exponentiated, provides the expected multiplicative change in turnout

rate in a no rapid transit division relative to a high rapid transit division. Similarly, the low transit coefficient (β_3), when exponentiated, provides the expected multiplicative change in turnout rate in a low rapid transit division relative to a high rapid transit division. When exponentiated, β_4 provides the additional multiplicative change in turnout rate associated with a one-unit increase in standardized poverty score in no-transit divisions, beyond the baseline poverty effect (β_1) already estimated for high-transit divisions. That is, the total poverty effect in no-transit divisions is given by exponentiating $\beta_1 + \beta_4$, while β_4 alone reflects how much the poverty-turnout relationship differs between no-transit and high-transit divisions. Likewise, exponentiating β_5 captures the additional poverty effect in low-transit divisions relative to high-transit divisions; the total poverty effect in low-transit divisions given by exponentiating $\beta_1 + \beta_5$.

This study uses a log-offset Poisson regression to capture the change in the rate of in-person election day turnout relative to the amount of registered voters in a given division. The amount of in-person election day votes a division receives cannot exceed the number of registered voters. A standard Poisson model assumes that the mean of the outcome (here votes) is equal to a variance; where they are not equal, overdispersion occurs. To account for overdispersion, I use a quasipoisson regression. Lastly, including a fixed-effects term in the equation helps mitigate year-specific election shocks. For example, in-person election day turnout can vary across elections as a result of other circumstances, like election saliency or weather. Altogether, a log-offset fixed-effects quasipoisson regression is the best approach to evaluate the central question because it models voter turnout as a rate and isolates the effect of rapid transit access from time-invariant division differences and year-specific election shocks.

Additionally, the analysis needed to account for spatial correlation. To test whether model residuals exhibited spatial dependence, I conducted a Moran's I test. The results revealed a positive spatial autocorrelation in the residuals (Moran's ($I = 0.56$), ($p < .001$)). Standard errors clustered by ward were used to calculate significance levels. The Moran's I test result serves as the motivation to implement a permutation test in future research, see more in the *Future Considerations* section.

6 Results

6.1 Descriptive Results

Figure 2 displays the standardized poverty score per division in Philadelphia. A higher poverty score is indicative of higher poverty rates in a division relative to the rest of the city, whereas a lower poverty score is indicative of lower poverty rates.

For the voting data, the study sample investigated in-person election day turnout across 1,703 divisions in Philadelphia for each fo the nine elections that occurred between May 2021 and May 2025 (five primary elections and four general elections). Figure 3 descriptively displays the turnout on election day in Philadelphia divisions from 2021-2025 as a percentage of registered voters. The baseline election day turnout in Philadelphia between 2021 and 2025 was 21.7%; in-person turnout has ranged from 11.44% in the May 2025 primary election to 46.62% in the November 2024 general election. The map displays a distinct spatial divide, with consistently high percentages of in-person election day voter turnout between 2021-2025 concentrated in far Northwest and Northeast Philadelphia, and consistently low turnout concentrated in North Philadelphia.

Before implementing the regression, I investigated the general trends of in-person election day turnout across

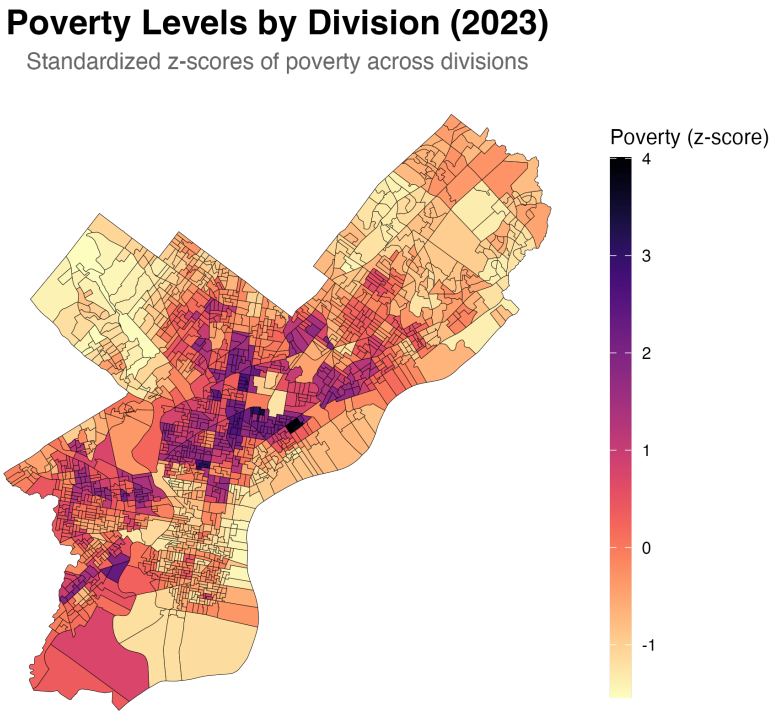


Figure 2: Standardized Poverty Z-Score by Division, 2023

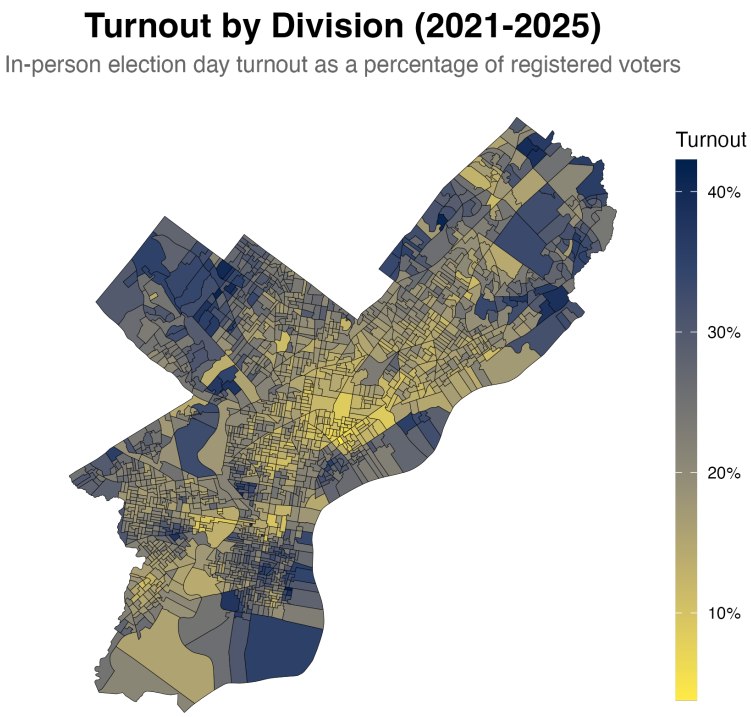


Figure 3: In-Person Turnout by Division, 2021-2025

divisions grouped by poverty-level and rapid transit availability. Figure 4 displays a descriptive statistics table that covers general turnout trends for divisions as defined by their standardized poverty scores and the level of availability of rapid transit.

Rapid Transit	Socioeconomic Status	# Precincts	Turnout %
No Rapid Transit	Wealthy	575	25.7
Low/Moderate Rapid Transit	Wealthy	8	22.2
No Rapid Transit	Average	562	21.0
High Rapid Transit	Wealthy	22	20.2
High Rapid Transit	Average	45	20.0
Low/Moderate Rapid Transit	Average	22	18.6
High Rapid Transit	Moderate Poverty	31	18.2
No Rapid Transit	Moderate Poverty	244	18.1
Low/Moderate Rapid Transit	Moderate Poverty	18	17.3
High Rapid Transit	High Poverty	12	17.1
No Rapid Transit	High Poverty	155	15.1
Low/Moderate Rapid Transit	High Poverty	9	13.6

Figure 4: Table of Voter Turnout by Transit and Poverty, Philadelphia 2021-2025

In this table, wealthy divisions are defined by a poverty score of -2 to -0.50, average divisions are defined by a poverty score of -0.50 to 0.50, moderate poverty divisions are defined by a poverty score of 0.50 to 1.50, and high poverty divisions are defined by a poverty score above 1.50. A score closer to 0 means that a division is closer to the citywide average.

The table indicates that wealthy divisions with no rapid transit and low-moderate rapid transit demonstrate the highest rates of election day turnout. On the opposite end of the spectrum, high poverty divisions with no rapid transit and low-moderate rapid transit demonstrate the lowest election day turnout during the same time period. However, within high and moderate poverty areas, those with high rapid transit access have the highest in-person election day turnout rates. Therefore, the results of this table provide further evidence that poverty may be a significant factor in the relationship between transit and turnout.

6.2 Regression Results

Upon investigating the relationship between rapid transit availability and in-person voter turnout in isolation between 2021 to 2025, there were varying effects across different classifications of rapid transit levels. Poverty emerged as a significant underlying factor. Higher poverty is associated with lower in-person election day vote rates in areas with high rapid transit access. A one-unit increase in standardized poverty score reduces the expected vote rate by approximately 4.1% holding other factors constant, see Figure 5 below.

As Figure 6 shows, in low-moderate rapid transit divisions, for every one-unit increase in standardized poverty score, there is an associated 8% decrease in expected in-person election day turnout in comparison to high rapid transit divisions. This effect is even stronger in divisions with no rapid transit. In divisions with no rapid transit, for every one-unit increase in standardized poverty score, there is a significant 13.6% decrease in expected in-person election day turnout in comparison to high rapid transit divisions. These findings provide evidence that areas with no rapid transit access see the largest drop in in-person election day turnout as poverty increases, suggesting that

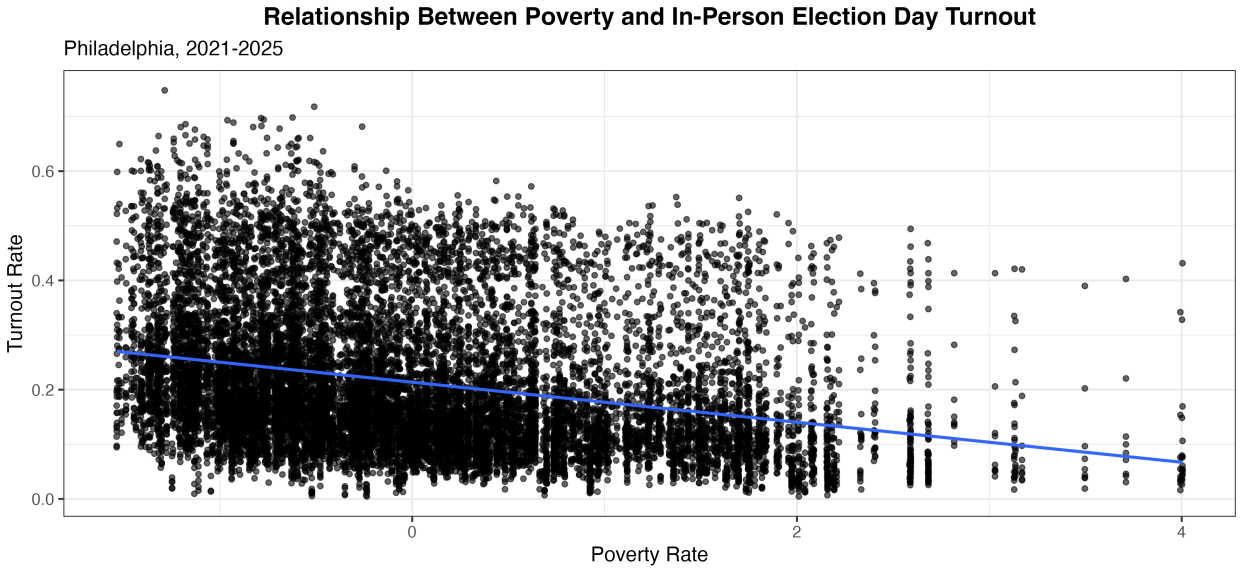


Figure 5: Relationship Between Poverty and In-Person Election Day Turnout, 2021-2025

improved rapid transit access may buffer the negative impact of poverty on turnout.

Term	Percent Change	95% CI	p value
Poverty	-4.13	(-10.39, 2.56)	0.220
Low/Moderate Transit	-2.22	(-12.78, 9.61)	0.699
No Transit	9.98	(2.56, 17.94)	0.008
May 2022	4.26	(3.43, 5.10)	<0.001
May 2023	32.02	(30.79, 33.26)	<0.001
May 2024	-17.09	(-18.07, -16.10)	<0.001
May 2025	-22.86	(-23.71, -22.00)	<0.001
November 2021	-4.08	(-4.91, -3.24)	<0.001
November 2022	125.47	(122.63, 128.35)	<0.001
November 2023	49.13	(47.77, 50.50)	<0.001
November 2024	204.10	(198.71, 209.59)	<0.001
Poverty × Low/Moderate Transit	-8.01	(-17.59, 2.67)	0.136
Poverty × No Transit	-13.58	(-19.32, -7.42)	<0.001

Figure 6: Regression Output (Percent change computed as $100\left(e^{\beta}-1\right)$) with Standard Errors Clustered by Ward

As mentioned before, the average voter turnout rate in Philadelphia between May 2021 and May 2025 across nine primary and general elections is 21.7%. The estimates from the regression show that for each unit increase in poverty, there is a 4.1% decrease in turnout. So, in practical terms, with a one-unit increase in poverty score, there will be approximately 20.8% turnout in a division with high rapid transit access, 19.1% turnout in a division with low-moderate rapid transit access, and 17.9% turnout in a division with no rapid transit access.

6.3 Spatial Autocorrelation

This findings further revealed that observations were not spatially independent and that unmeasured neighborhood processes may also influence in-person voter turnout. I estimated cluster-robust standard errors at the ward level using heteroskedasticity-consistent variance estimators for the regression coefficients. After applying clustered standard errors, the interaction between poverty and no rapid transit access remained statistically significant. This provides additional evidence that the negative association between poverty and in-person voter turnout was substantially stronger in areas without rapid transit infrastructure.

7 Discussion

7.1 Limitations

It is important to acknowledge limitations in the methodological choices and available data for this study. First, this study only analyzes the relationship between transit and turnout between 2021 and 2025, capturing short-term trends in this relationship. Furthermore, the operationalization of rapid transit relies on simple counts of rapid transit stops in each division, as opposed to other metrics like cost and/or frequency. Thus, the measure of rapid transit as an independent variable may not capture the full range of complexities of how an individual's means of transportation factors in as a cost of voting. Additionally, this study does not identify causal mechanisms for how rapid transit in impoverished areas is associated with increased turnout. Future research could assess possible causal mechanisms.

7.2 Conclusion

This study explored the relationship between access to rapid transit and its impact on in-person election day turnout in Philadelphia from 2021-2025. In addition, this study analyzes how socioeconomic status further influences this relationship. I find that in divisions with higher rates of poverty, greater access to rapid public transit is associated with higher levels of in-person election day turnout in comparison to divisions without rapid transit access. In other words, these findings suggest that accessible rapid transit may mitigate the negative effects of poverty on in-person election day turnout.

These findings align with the Rational Voter Theory and other theories of spatial inequality, suggesting that physical accessibility is a key component of political participation. For residents in high-poverty areas, limited transit options can create significant logistical barriers to voting, including time, lost wages, and childcare costs associated with voting. Improved rapid transit networks may lower the perceived cost of voting modeled in Rational Voter Theory.

Ultimately, the findings of this study provide strong evidence for how increased investment and expansion of public transportation systems in Philadelphia can increase in-person election day voter turnout in some of the city's most vulnerable communities. There is ample evidence that public transportation systems are critical in providing social and economic mobility for urban residents, and now there is more evidence that the importance of public transportation also extends to political participation. The continued disinvestment in public transportation poses serious consequences for democratic participation and thereby political representation of impoverished and underrepresented communities in Philadelphia. Spatial inequalities have cemented their effects in the foundation of

the built environment of Philadelphia. In turn, solutions to these systems of inequality that perpetuate disparities in democratic participation require actions that reshape how public transportation infrastructure facilitates mobility in multiple aspects — physically, economically, and, as this study suggests, politically. The significance of this work manifests in three key ways:

1.) **Political representation of vulnerable groups.** This research highlights how disparities in voter turnout are reinforced through a self-perpetuating feedback loop, as inadequate public infrastructure creates barriers to political participation. Consequently, these barriers restrict individuals' ability to elect representatives who may advocate for policies that improve those very systems.

2.) **Urban democratic participation in swing states.** This research sheds light on the broader electoral implications of suppressed urban democratic participation. Because large urban populations are important in shaping electoral outcomes, especially in swing states, barriers to democratic participation produce ripple effects that can extend beyond local areas to influence state and national election outcomes.

3.) **Social mobility.** This research provides further evidence to the argument that public transportation is a tool for social mobility. Existing literature shows that insufficient access to public transportation constrains educational and employment opportunities. Building on this understanding, this study suggests that limited public transportation access decreases democratic participation, particularly in lower-income communities, where reduced political representation further restricts the ability to advocate for policies and investments that promote upward mobility.

7.3 Future Considerations

Moran's I testing revealed significant positive spatial autocorrelation. Thus, unmeasured spatially clustered factors, such as local organizing, may influence turnout in ways not captured by the current model. To partially address this, cluster-robust standard errors were applied at the ward level. However, this correction does not fully resolve spatial dependence across ward boundaries, and this study will, in the future, use methods such as permutation tests to more completely account for this structure.

Future research should disaggregate turnout by election type, primary and general elections. Furthermore, the association between rapid transit access and mail-in ballots, rather than in-person votes, should be assessed. If similar results are observed for mail-in ballots, this may suggest that transit facilitates democratic participation more generally.

Although this analysis investigates this relationship in the context of Philadelphia, the results may differ across geographies depending on types of governance structures, disparities in poverty, and transportation planning practices. Future research should assess how race also affects the relationship between rapid transit and in-person voter turnout. Although the practice of redlining has been outlawed since the late 1960s, the harmful effects of its racial discrimination persist today in many cities, including Philadelphia. To account for how variance in governance structure and disparities in poverty in a given city factor into the relationship between transit and turnout, this study could be replicated in other largely populated U.S. cities that vary in historical context, geography, and racial composition.

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